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Course Code: DSA0401

Course Name: FUNDAMENTAL OF DATASCIENCE

Question 6: Grocery Bill with Discount and Tax

AIM:

To calculate the total cost of grocery items, including discount and tax.

ALGORITHM:

1. Store item prices and quantities in lists.

2. Calculate the subtotal.

3. Calculate the discount amount.

4. Subtract the discount and calculate tax.

5. Display the final bill.

CODE:

```
prices = [100, 200, 150]
quantities = [2, 1, 3]
discount_rate = 10
tax_rate = 5

subtotal = sum(p * q for p, q in zip(prices, quantities))
discount = subtotal * discount_rate / 100
after_discount = subtotal - discount
tax = after_discount * tax_rate / 100
```

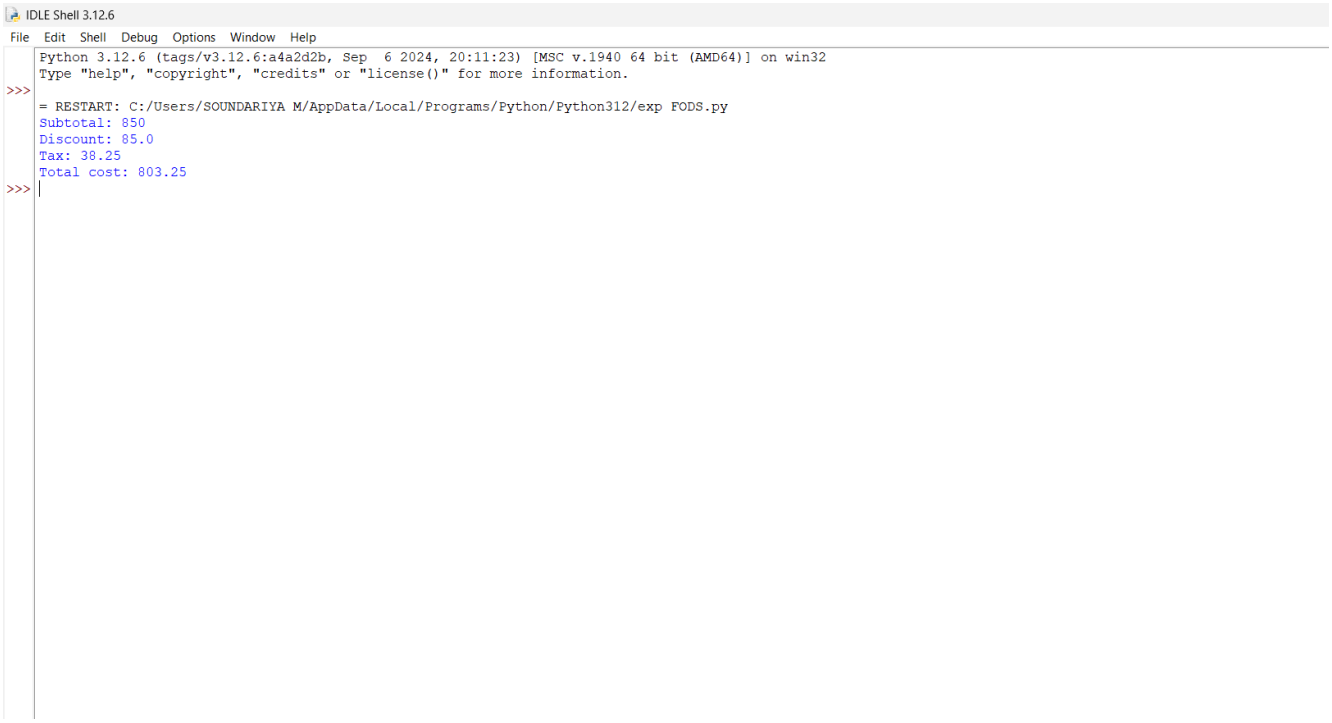
```
total = after_discount + tax

print("Subtotal:", subtotal)
print("Discount:", discount)
print("Tax:", tax)
print("Total cost:", total)
```

INPUT:

Prices = [100, 200, 150]; Quantities = [2, 1, 3]; Discount = 10%; Tax = 5%

OUTPUT:



RESULT:

The final grocery bill after discount and tax is ₹803.25.